Bernoulli extremizers for the independent Berry–Esseen ratio

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Abstract

We study the Berry–Esseen ratio of independent centered two-point variables, allowing zero summands. At every fixed number of available coordinates, every maximizing array with a common support diameter has identically distributed nonzero coordinates. With at most two coordinates, the same conclusion holds for arbitrary support diameters, and only one coordinate is nonzero. A uniform binomial phase expansion determines the asymptotic choice of active count when zeros are allowed: eventually at most four zeros are needed, and their frequencies are explicit. For binary arrays with ratios bounded away from zero, large total count variance and small diameter deviations, replacement by a common diameter strictly increases the ratio. The gain is quantified without an individual probability floor, and improves when the probabilities stay in a compact subset of $(0, 1)$. The finite classifications use interval certificates with explicitly stated domains. The replacement argument is analytic.

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1 The extremal questions

Let $X_1, \dots, X_n$ be independent centered real variables and put

$$V = \sum_i \mathbb{E}X_i^2 > 0, \qquad R = \sum_i \mathbb{E}|X_i|^3 < \infty, \qquad \mathcal{C}(X) = \frac{V^{3/2}}{R} \sup_y \left| \mathbb{P}\{\textstyle\sum_i X_i \le y\} - \Phi(y/\sqrt{V}) \right|.$$

Here Φ and ϕ are the standard normal distribution function and density. A binary coordinate has the form $h_i(B_i - p_i)$, where $h_i > 0$ and $B_i \sim \mathrm{Bern}(p_i)$. Its support diameter is h_i. Parameters $p_i = 0, 1$ give a centered zero coordinate. For binary coordinates,

$$V = \sum_i h_i^2 p_i(1-p_i), \qquad R = \sum_i h_i^3 p_i(1-p_i)\{p_i^2 + (1-p_i)^2\}. \tag{1.1}$$

The conjectured unrestricted bound is

$$\mathcal{C}(X) \le C_{\mathrm{E}} := \frac{3+\sqrt{10}}{6\sqrt{2\pi}}.$$

Esseen's fixed-law asymptotics give this value as a necessary lower bound, using repeated copies of a binary law with upper mass $p_{\mathrm{E}} = (4-\sqrt{10})/2$ [4, pp. 160–163]. Schulz proves the sharp bound for binomial sums [8, Theorem 1]; a companion paper proves it for independent binary coordinates having a common diameter [1]. The present paper identifies the finite maximizers and proves replacement theorems beyond exact diameter equality. It also explains the zero padding visible in finite binomial optimization. The integer threshold cannot remain at its preferred phase while the probability remains at its asymptotic optimum; a smaller active core can have a better phase and a larger ratio.

There are two separate difficulties. Unequal masses change both the variance and the third-moment denominator. Unequal diameters can split a convolution atom, so the objective need not be differentiable under support motions. Our finite common-diameter argument varies probabilities at a fixed integer threshold. Our diameter comparison replaces the whole diameter vector and keeps the exact probabilities of crossing each count layer.

Let B_m be the maximal ratio for m identically distributed active binary coordinates. Let C_n^{com} be the maximum over at most n common-diameter binary coordinates. The maximum allows zero coordinates. Let C_1 be the one-coordinate binary maximum; explicitly,

$$C_1 = \max_{0<p<1} \frac{\sqrt{p(1-p)}}{p^2+(1-p)^2} \left\{ \Phi\left(\sqrt{\frac{p}{1-p}} \right) - p \right\}. \tag{1.2}$$

Bentkus and Kirsa prove that this is also the unrestricted one-coordinate constant [2, Theorem 1].

Theorem 1.1 (Common diameter)**.** *For every $n \ge 1$,*

$$C_n^{\mathrm{com}} = \max_{1 \le m \le n} B_m.$$

Every maximizing array consists of identically distributed active binary coordinates and zero coordinates.

Theorem 1.2 (Two arbitrary binary diameters)**.** *Over arrays with at most two binary coordinates, the maximal ratio is C_1. Every maximizing array has exactly one active coordinate. If both variances must be positive, C_1 is an unattained supremum.*

Theorems 1.1 and 1.2 include finite interval calculations. Their precise arithmetic premises and the analytic boundary arguments are given in Sections 2 and 3. The records and verifier are described in `certificates/structure/README.md` and `verify/structure/`. These finite classifications do not assert uniqueness of the maximizing active count or probability. The eventual binomial phase results are given in Section 4.

Theorem 1.3 (Small diameter spread)**.** *Fix $c_0 > 0$. There are $A_0, \sigma_0, \eta_0, \kappa > 0$, depending only on c_0, with the following property. Let $X_i = g_i(B_i - p_i)$, where $g_i > 0$ and $0 < p_i < 1$. Put*

$$a_i = p_i(1-p_i), \quad A = \sum_i a_i, \quad h_0 = \frac{\sum_i a_i g_i}{A}, \quad e_i = g_i/h_0 - 1, \quad \sigma^2 = \sum_i a_i e_i^2, \quad \eta = \max_i |e_i|.$$

If $A \geq A_0$, $0 < \sigma \leq \sigma_0$, $\eta \leq \eta_0$ and $\mathcal{C}(X) \geq c_0$, then

$$\mathcal{C}(B_1 - p_1, \ldots, B_n - p_n) - \mathcal{C}(X) \geq \kappa \frac{\sigma^2}{\sigma + \eta} > 0.$$

The comparator preserves every individual probability. If all probabilities belong to a fixed compact $J \subset (0,1)$, the gain is at least $\kappa_J \sigma$ for some $\kappa_J > 0$.

The maximum deviation condition controls diameters carried by arbitrarily small probabilities. The constants in Theorem 1.3 are qualitative. Section 5 also gives a competitive binary family for which the gain has exactly the smaller order $\sigma^2/(\sigma + \eta)$.

For every finite-support sum we use both atom sides:

$$\sup_y |F(y) - \Phi(y/\sqrt{V})| = \sup_y \max\{F(y) - \Phi(y/\sqrt{V}),\, \Phi(y/\sqrt{V}) - F^-(y)\}, \tag{1.3}$$

where $F(y) = \mathbb{P}(S \leq y)$ and $F^-(y) = \mathbb{P}(S < y)$. Reflection interchanges the branches. Between consecutive atoms, the positive closed discrepancy decreases. Thus its maximum occurs at an atom, and one selected positive closed threshold suffices after reflection, including ties.

2 All finite common-diameter maximizers

Scale the common diameter to one and write

$$\mu = \sum_i p_i, \quad V = \sum_i p_i(1-p_i), \quad R = \sum_i \{p_i - 3p_i^2 + 4p_i^3 - 2p_i^4\}, \quad F_k = \mathbb{P}\{\textstyle\sum_i B_i \leq k\}.$$

Since $R \geq V/2$, the ratio is at most $2\sqrt{V}$. Its extension by zero at $V = 0$ is continuous on the compact probability cube. Thus $C = C_n^{\mathrm{com}}$ is attained. At any selected closed integer threshold, define

$$H_C(p) = \frac{CR}{V^{3/2}} + \Phi((k-\mu)/\sqrt{V}) - F_k. \tag{2.1}$$

The global definition of C gives $H_C \geq 0$ everywhere with $V > 0$, and equality at every maximizing array at its selected threshold.

The probability replacements below have classical precedents. Hoeffding's pair identity preserves the sum of two probabilities [5, equation (10), pp. 714–716]. Mattner's few-moment extremum theorem and next-power-sum argument [7, Theorems 1.1–1.2 and Lemma 2.1] use related fibers for separately affine symmetric functions. The present deficit contains a cubic-moment denominator. We establish its required affinity directly.

2.1 A uniform third-difference estimate

For the mass function c_j of a Poisson-binomial sum, extended by zero, put $\Delta^3 c_j = c_j - 3c_{j-1} + 3c_{j-2} - c_{j-3}$.

Lemma 2.1. *If that sum has variance W, then*

$$\sup_j |\Delta^3 c_j| < \frac{16/5}{(1+W)^{3/2}}.$$

Proof. For $W \le 1/4$, choose the more likely outcome of each bit as a base configuration. Let J count departures, $\epsilon_i = \min(p_i, 1-p_i)$, $E = \sum \epsilon_i$, and $Q = \sum \epsilon_i/(1-\epsilon_i)$. Then $E \le 2W \le 1/2$, $W \le Q \le 1$,

$$\mathbb{P}(J=0) \ge e^{-Q}, \qquad \mathbb{P}(J=1) = Q\mathbb{P}(J=0) \ge We^{-W}.$$

At the two indices where the base state contributes magnitude three, the one-departure terms have the opposite sign and magnitude at least one. The base probability is at least $1/2$, so the sign cannot reverse. Thus the absolute third difference is at most $3 - 4We^{-W}$ there. At all other indices it is at most $1 + 4W$. For the central bound use $e^{-W} \ge 1 - W$ and

$$(1+W)^{-3/2} \ge h(W) := 1 - 3W/2 + 15W^2/8 - 35W^3/16.$$

The polynomial $(16/5)h(W) - (3 - 4W + 4W^2)$ is decreasing on $[0, 1/4]$: its derivative is bounded above by $-1/2 + 13W/4 - 315W^2/16 < 0$. At $W = 1/4$ it equals $1/64$. The outer estimate is at most two, below $(16/5)(5/4)^{-3/2}$.

For $W \ge 1/4$, Fourier inversion gives

$$|\Delta^3 c_j| \le I(W) := \frac{16}{\pi} \int_0^{\pi/2} \sin^3 u \, e^{-2W \sin^2 u} \, du.$$

Write $J_r = \int_0^{\pi/2} \sin^r u \, e^{-2W \sin^2 u} \, du$. Integration of $(\sin^4 u \cos u \, e^{-2W \sin^2 u})'$ gives $(5 + 4W)J_5 = 4J_3 + 4WJ_7$. Consequently

$$\frac{d}{dW} \log\{I(W)(1+W)^{3/2}\} \le -\frac{8}{5+4W} + \frac{3}{2(1+W)} < 0.$$

At $W = 1/4$, the exponential upper Taylor polynomial through degree six and the exact odd sine integrals give

$$I(1/4)(5/4)^{3/2} \le \frac{10\sqrt{5}}{\pi} \, \frac{911264}{2027025} < \frac{16}{5}.$$

The last strict inequality follows by squaring and using $\pi > 6283/2000$; it is one of the rational scalar controls in the computational supplement. □

2.2 Reducing the number of distinct probabilities

Fix a maximizing array and its mean, variance and closed threshold. Consider the compact zero set

$$Z = \{p \in [0,1]^n : \ \sum p_i = \mu, \ \sum p_i(1-p_i) = V, \ H_C(p) = 0\}.$$

Let K be its connected component containing the array, and maximize $\sum p_i^3$ on K. For three selected probabilities, let e_1, e_2, e_3 be their elementary symmetric functions. Their generating polynomial is

$$1 + e_1(w-1) + e_2(w-1)^2 + e_3(w-1)^3.$$

Holding e_1, e_2 fixed preserves their mean and variance. Their contribution to R is

$$R_3 = e_1 - 3e_1^2 + 4e_1^3 - 2e_1^4 + (6 - 12e_1 + 8e_1^2)e_2 - 4e_2^2 + 4(3 - 2e_1)e_3.$$

Both F_k and R are affine in e_3 on this fiber. If its three roots are distinct and interior, e_3 can move in both directions. A nonzero slope contradicts $H_C \geq 0$. A zero slope gives a continuous path in Z, hence in K, on which $\sum p_i^3 = e_1^3 - 3e_1e_2 + 3e_3$ increases. The chosen point therefore has at most two distinct active probabilities.

Suppose they are $a < b$, each occurring at least twice. The triples (a, a, b) and (a, b, b) are opposite endpoints of their three-root fibers. If d_j is the mass function of the other coordinates, the fiber slope is

$$L = \frac{4C(3 - 2e_1)}{V^{3/2}} + \Delta^2 d_k.$$

The two endpoint conditions give $L_{aab} \leq 0 \leq L_{abb}$. Delete two bits of each type, let c_j be the remaining mass function, and let W be its variance. Direct subtraction gives

$$L_{abb} - L_{aab} = (b - a)\left\{\Delta^3 c_k - \frac{8C}{V^{3/2}}\right\}.$$

Since $V \leq W + 1$, Lemma 2.1 contradicts these signs whenever $C \geq 2/5$. Thus one of the two multiplicities is one. At $n = 3$ this is automatic. At $n = 4$, the only remaining repeated pattern is $(2, 2)$; its deleted law is deterministic, so $|\Delta^3 c_k| \leq 3$, whereas $8C/V^{3/2} \geq 8C > 3$ when $C \geq 199/500$.

2.3 Strict curvature of the remaining fiber

The remaining unequal pattern is $\mathrm{Bin}(M, p) + \mathrm{Bern}(x)$, $p \neq x$. We record its exact curvature and the finite inequality used to exclude it. Let $q = 1-p$, $m = M-1$, $d_j = \mathbb{P}(\mathrm{Bin}(m, p) = j)$, $W = mpq$, and

$$V = x(1-x)+Mpq, \quad z = \frac{k - x - Mp}{\sqrt{V}}, \quad \xi = d_{k-1}-d_k, \quad S_2 = (1-2x)^2+(1-2x)(1-2p)+(1-2p)^2.$$

Preserve the sum of the exceptional mass and one mass p, write them as $\bar{p}+d, \bar{p}-d$, and set $u = d^2$. Along this path,

$$F_k' = \xi, \qquad V' = -2, \qquad R' = -2S_2, \qquad R'' = -8.$$

At stationarity of H_C, differentiating its smooth terms and eliminating R gives

$$V^2 H_C'' = 5V\xi - C\left(8\sqrt{V} + \frac{2S_2}{\sqrt{V}}\right) - z\phi(z)(2 + z^2). \tag{2.2}$$

A minimum requires this expression to be nonnegative.

Lemma 2.2 (The certified curvature inequality)**.** *Put $c_* = 199/500$. For*

$$0 < p, x < 1, \quad pq \leq W \leq 3, \quad 0 \leq f \leq 1, \quad A_0 = W - (1 - f)q^2 \geq 0,$$

define

$$B_0 = p^2 + (1 - f)(W + 2pq), \quad V = x(1 - x) + W + pq, \quad z = \frac{f(W + pq)/(1 - qf) - x}{\sqrt{V}}.$$

Then

$$\frac{2c_*}{\sqrt{V}}\{4W + 2 + (1 - 2x)(1 - 2p)\} + z\phi(z)(2 + z^2) - \frac{5VfB_0}{(2 - f)B_0 + A_0} > 0. \tag{2.3}$$

Also every two-bit unit-diameter array has ratio less than c_*.

The lemma is a finite arithmetic premise. Its compact interval cover at 192 Arb bits [6] processed 550563 boxes for (2.3) and 13641 boxes for the two-bit assertion, with no unresolved box. The archived records bind the evaluated formulas and the declared domains. Below we explain the covering charts and the analytic complements; rebuilding those interval covers is distinct from checking the saved hashes and coverage metadata.

For $V \le 1/5$, the first term in (2.3) is at least $8c_*/\sqrt{3}$, the normal term is at least $-4/(e\sqrt{\pi})$, and the final term is at most one. Rational controls give

$$\frac{8c_*}{\sqrt{3}} - \frac{4}{e\sqrt{\pi}} - 1 > \frac{1}{247}, \quad 8c_*\sqrt{3} - \frac{10}{\pi} - \frac{4}{e\sqrt{\pi}} > 1, \quad 8c_*\sqrt{7/2} - \frac{35}{3\pi} - \frac{4}{e\sqrt{\pi}} > \frac{44}{45}. \tag{2.4}$$

For $V \ge 1/5$ use $z\phi(z)(2+z^2) \ge \sqrt{2/\pi}\min(z,0)$. The following maps from $[0,1]^4$, with fourth variable x, cover the entire remaining domain:

$$\begin{aligned} &p = (1+s_p)/2, \quad W = pq + s_W(3-pq), \quad f = s_f; \\ &p = s_p/2, \quad W = q^2 + s_W(3-q^2), \quad f = s_f; \\ &p = s_p/2, \quad W = pq + s_W(q^2-pq), \quad f = 1-(1-s_f)W/q^2. \end{aligned}$$

The first covers $p \ge 1/2$. The last two split $p \le 1/2$ at $W = q^2$; below that interface the last map imposes exactly $f \ge 1 - W/q^2$. Interval division is used only with a strictly positive denominator. Otherwise the universal neighbor-fraction bound one, or $y \ge 0$ at a singular endpoint, supplies a valid lower bound. A box is accepted only when its outward lower endpoint is positive. The two-bit cover includes the entire probability square and all three closed integer thresholds. The zero-variance corners use $\mathcal{C} \le 2\sqrt{V}$ before division.

To apply the lemma to (2.2), first suppose $M \ge 2$. If $\xi \le 0$, then $\max(V, S_2) \ge 1/3$ and

$$C(8\sqrt{V} + 2S_2/\sqrt{V}) + z\phi(z)(2+z^2) \ge 8c_*/\sqrt{3} - 4/(e\sqrt{\pi}) > 0.$$

Thus $H_C'' < 0$. If $\xi > 0$, binomial recurrence gives

$$y = k - (m+1)p > 0, \quad f = y/(kq) \in (0,1], \quad d_k = (1-f)d_{k-1}, \quad d_{k-2} = r_0 d_{k-1},$$

where

$$r_0 = A_0/B_0, \qquad y = f(W+pq)/(1-qf).$$

The sum of these three masses is at most one, so $\xi \le f/(2-f+r_0)$. For $W \le 3$, Lemma 2.2 therefore makes (2.2) negative. The coefficient of C is positive, so the same conclusion holds for every actual extremal ratio $C \ge c_*$. For $W \ge 3$, Fourier inversion gives

$$|\xi| \le \frac{4}{\pi}\int_0^{\pi/2} \sin u\, e^{-2W\sin^2 u}\, du < \frac{2}{\pi W}.$$

The negative of the right side of (2.2) is at least $8c_*\sqrt{V} - 4/(e\sqrt{\pi}) - 10V/(\pi W)$. It is concave in $V \in [W, W+1/2]$. Both endpoint values increase with W, and (2.4) gives positive bounds at $W = 3$. For $M = 1$ the two-bit assertion of the lemma excludes the pattern at price $C \ge c_*$.

2.4 From one equal-mass point to every maximizer

The cubic maximizer on K is now an IID group plus deterministic bits. The number m of active bits is at least three, by Lemma 2.2. We prove this point is isolated in Z. For a pair in the IID group, use $p+d, p-d$ and $u = d^2$. Nonnegativity gives $\Lambda := \partial_u H_C(0) \ge 0$. If $\Lambda = 0$, (2.2) applies

at the diagonal with deleted law $\mathrm{Bin}(m-2,p)$. The compact lemma or its analytic complement makes $\partial_u^2 H_C(0) < 0$, contradicting $H_C \geq 0$ for small positive u. Hence $\Lambda > 0$. Permutation symmetry makes the exchange quadratic form $(\Lambda/2)\|y_{\mathrm{cen}}\|^2$ on the mean-zero directions of the active probabilities.

For a nearby point in Z, let $a_i \geq 0$ activate zero probabilities, $b_j \geq 0$ deactivate probability-one coordinates, and y_l change the m active probabilities. Put $a = \sum a_i$, $b = \sum b_j$, $t = a + b$. The two constraints give exactly

$$\sum_l y_l = b - a, \qquad \sum_l y_l^2 = 2pa + 2(1-p)b - \sum_i a_i^2 - \sum_j b_j^2.$$

Thus $\|y_{\mathrm{cen}}\|^2 \geq c_p t - O(t^2)$, with $c_p > 0$. The endpoint linear coefficients of H_C are nonnegative: activate one endpoint and compensate uniformly in the active group to preserve the mean, then apply $H_C \geq 0$ on that feasible path. The Taylor expansion on the fixed mean/variance fiber consequently has nonnegative endpoint linear terms, the positive exchange term, and remaining terms $O(t^{3/2})$. The uniform active direction is $O(t)$, the exchange direction is $O(\sqrt{t})$, so these remainder orders also include their mixed terms. For small $t > 0$ this gives $H_C > 0$. If $t = 0$, the two constraints force $y = 0$. The IID point is isolated. Its connected component K is therefore a singleton, and the original maximizer was the same IID group plus zeros.

The required lower prices follow from two exact admissible examples:

active count	p	closed raw threshold	ratio lower bound
3	43/100	1	$0.3981444741 > 199/500$
5	9/20	2	$0.4009890598 > 2/5$.

Their binomial probabilities are rational. Rational square-root bounds, Machin bounds for π, and the integrated Gaussian Taylor polynomial $x - x^3/6 + x^5/40 - x^7/336$ give the displayed strict lower bounds. Padding preserves them. The three-bit example handles $n = 3, 4$; the five-bit example supplies the $2/5$ hypothesis for the repeated-type argument at every $n \geq 5$.

2.5 The remaining two-coordinate case

For two probabilities p, x, set $s = p + x$, $w = px$ and

$$V = s - s^2 + 2w, \quad R = s - 3s^2 + 4s^3 - 2s^4 + (6 - 12s + 8s^2)w - 4w^2.$$

At rank $k = 0, 1, 2$, the closed CDFs are $1 - s + w$, $1 - w$, 1. Let $z = (k - s)/\sqrt{V}$, $D = F_k - \Phi(z)$ and $C = DV^{3/2}/R$. Off the diagonal the map $(p, x) \mapsto (s, w)$ is invertible, so an interior local maximum requires both derivatives to vanish. With $F_w = (1, -1, 0)_k$, $F_s = (-1, 0, 0)_k$, $V_s = 1 - 2s$, $R_w = 6 - 12s + 8s^2 - 8w$ and $R_s = (1 - 2s)^3 + (-12 + 16s)w$, their positive-denominator numerators are

$$\begin{aligned} G_w &= R(VF_w + z\phi(z) + 3D) - DVR_w, \\ G_s &= 2VRF_s + 2\sqrt{V}R\phi(z) + Rz\phi(z)V_s + 3DRV_s - 2DVR_s. \end{aligned}$$

At $G_w = 0$,

$$\frac{C_{ww}}{C} = \frac{N}{DV^2R}, \quad N = R\{6F_wV + z\phi(z)(3 + z^2) + 3D\} + 8DV^2.$$

The finite critical certificate covers each owner square by $C < 37/100$, $0 \notin G_w$, $0 \notin G_s$, or $N > 0$. A separate diagonal cover uses only the ratio bound, since the (s, w) chart is singular there. These

covers processed respectively 4683 and 195 boxes, with no unresolved box. They exclude every two-active-bit common-diameter maximizer at ratio at least 37/100. Boundary derivatives are not assumed to vanish; the cover bounds their limits only to enclose the open stationary domain. The exact one-bit example $p = 2/5$ gives a ratio greater than 37/100. Compact attainment therefore leaves exactly one active bit at $n = 2$. The $n = 1$ case is immediate. This completes Theorem 1.1.

3 Two arbitrary binary diameters

We first justify attainment and separate the boundary cases from the open stationary domain. Normalize $V = 1$ and take a maximizing sequence with ratio at least $c > 0$. For two coordinates, $R \geq 1/\sqrt{2}$, whereas the discrepancy is at most one, so $R \leq 1/c$. The squares are uniformly integrable, since

$$\sum_i \mathbb{E}[X_i^2; |X_i| > L] \leq \frac{1}{cL}.$$

After reflection, selected positive closed thresholds stay bounded: Chebyshev controls thresholds tending to $-\infty$, and $1 - \Phi(y)$ controls those tending to $+\infty$. Coordinatewise weak limits preserve centering and total variance. A weak limit of two-point laws has at most two points. Closed-set Portmanteau at the moving thresholds and lower semicontinuity of R give a limiting ratio at least the supremum; equality follows from its definition. Thus a binary-class maximizer exists.

3.1 The top threshold and equal-diameter boundary

For a centered binary variable on $\{-b, a\}$, $a, b > 0$,

$$v = ab, \qquad \rho = \frac{ab(a^2 + b^2)}{a + b} \geq 2(\sqrt{2} - 1)\frac{v^2}{a}.$$

The inequality is the minimum of $(1 + x^2)/(1 + x)$ for $x = a/b > 0$. For two such coordinates with total upper endpoint L, Cauchy's inequality gives $R \geq 2(\sqrt{2}-1)V^2/L$. At the top closed threshold the CDF is one, so its ratio is at most

$$\frac{z\{1 - \Phi(z)\}}{2(\sqrt{2} - 1)} \leq \frac{e^{-1/2}}{4(\sqrt{2} - 1)} < \frac{37}{100}, \qquad z = L/\sqrt{V}.$$

Here $1 - \Phi(z) \leq e^{-z^2/2}/2$ follows by translating its Gaussian integral. For the final rational comparison, $\sqrt{e} > 41/25$, $\sqrt{2} - 1 > 207/500$ give $12500/33948 < 37/100$. Reflection treats the negative strict bottom threshold. The common-diameter two-active-bit case was excluded in Section 2. A zero coordinate leaves exactly the one-coordinate problem, whose ratio exceeds 37/100.

3.2 Open unequal-diameter stationary points

Scale the larger diameter to one and write the other as $t \in (0, 1)$. Let their upper probabilities be $p, q \in (0, 1)$, and put

$$a = p(1-p), \quad b = q(1-q), \quad r_p = p^2+(1-p)^2, \quad r_q = q^2+(1-q)^2, \quad V = a+t^2b, \quad R = ar_p+t^3br_q.$$

The raw support is $0, t, 1, 1 + t$. The three remaining positive closed thresholds have the data

raw threshold	F	y (centered threshold)
0	$(1 - p)(1 - q)$	$-p - tq$
t	$1 - p$	$-p + t(1 - q)$
1	$1 - pq$	$1 - p - tq.$

Set $z = y/\sqrt{V}$, $D = F - \Phi(z)$ and $C = DV^{3/2}/R$. At a global maximum, a selected active branch has a local maximum even if other thresholds tie. Its three derivatives therefore vanish. Define

$$\alpha = \phi(z)/\sqrt{V}, \quad A = \frac{3D + z\phi(z)}{2V}, \quad \lambda = D/R.$$

After removing the common positive factor $V^{3/2}/R$, the two probability derivatives are

$$g_p = F_p + \alpha + A(1-2p) - \lambda(1-2p)^3,$$
$$g_q = F_q + \alpha t + At^2(1-2q) - \lambda t^3(1-2q)^3.$$

Divide the diameter derivative by q for thresholds $0, 1$ and by $1-q$ for threshold t, then multiply by $VR\sqrt{V}$. Its numerator is

$$G = \phi(z)RB + 3Dwta\sqrt{V}\,(r_p - tr_q), \tag{3.1}$$

where the pairs (w, B) for these three rows are, respectively,

$$(1-q,\ p[(1-p) - t(1-q)]), \quad (q, -p[(1-p) + qt]), \quad (1-q, (1-p)[p + t(1-q)]).$$

The extra factors p and $1-p$ in the first and last rows are strictly positive in the open chart and can also be removed. These factorizations retain cancellation near vanishing probabilities.

3.3 A compact domain for the necessary equations

The probability equations also bound the normalized support points. For one centered binary coordinate with law P_i, put $s_i = \mathbb{E}X_i^2$, $b_i = \mathbb{E}|X_i|^3$, $M_i = \mathbb{E}X_i|X_i|$ and let G_i be the deleted CDF. In total-variance-one units define

$$Q_i(u) = G_i(y-u) - F(y) + \phi(y)u + A(u^2 - s_i) - C\{|u|^3 - b_i - 3M_iu\}.$$

Its mean under P_i is zero. Probability stationarity says its two endpoint values agree, so both equal zero. This uses probability stationarity alone, without a full-law majorant. Since $Cb_i \le CR = D \le 1$, $|M_i| \le s_i \le 1$, $\phi < 2/5$ and $|A| \le 13/8$, every active endpoint satisfies

$$C|u|^3 \le 4 + (2/5 + 3C)|u| + (13/8)u^2.$$

For $|u| \ge 3$, dividing the right side by u^2 bounds it by $C + 793/360$. In particular $|u| < 3 + 8/C$. At $C \ge 37/100$, the larger normalized diameter $1/\sqrt{V}$ is less than twice $911/37$. Hence

$$V > \frac{1369}{3319684} > \frac{1}{2500}, \qquad R \ge V^{3/2}/\sqrt{2} > V/75. \tag{3.2}$$

The arithmetic certificate covers the three closed cubes $(p, q, t) \in [0,1]^3$, conditional on (3.2) and $C \ge 37/100$. Each box is excluded by an empty conditional domain, a strict ratio upper bound, or a nonzero enclosure of a required probability or factored diameter derivative. One useful ratio bound is

$$C \le D\sqrt{a/r_p^2 + b/r_q^2},$$

which follows from Hölder's inequality applied to V, R. There are 37287 accepted source leaves in the complete partition. An independently written, cancellation-preserving replay checks every source condition, using 37421 descendant leaves and no unresolved or unstarted region. The input

bindings and exact binary forests are part of the public certificate. Boundary boxes enclose limits of open stationary points; they do not impose stationarity on actual one-bit boundary maximizers.

Consequently no competitive unequal-diameter two-active-bit maximum can satisfy its necessary equations. The preceding endpoint and common-diameter arguments cover all remaining two-active-bit cases. Attainment leaves exactly one active coordinate, proving Theorem 1.2. To require both variances positive, add a centered symmetric binary coordinate of diameter tending to zero. Choose the closed threshold just above the winning one-coordinate atom. Its CDF, variance and third-moment denominator converge to the one-coordinate values. Thus the supremum remains C_1, and the classification prevents its attainment with both coordinates active.

4 Binomial phase and the optimal number of zero coordinates

This section concerns the binomial optima B_m and their padded envelope $P_N := \max_{1\le m\le N} B_m$. It determines the eventual record pattern within this family. Theorem 1.1 transfers these conclusions to common-diameter maximizers; no reduction from arbitrary independent laws is assumed here.

Put $s = \sqrt{10}$ and write

$$\begin{gathered} p = (4-s)/2, \quad q = 1-p, \quad t = s-3, \quad u = p-t, \quad c_* = (20+s)/72, \\ Q_* = \frac{\phi(0)(867+287s)}{25920}, \qquad \alpha = \frac{\phi(0)(60+19s)}{30}, \qquad b = \frac{1010-397s}{12960}, \\ \delta = c_* - b = \frac{2590+577s}{12960}, \qquad \eta = Q_*/\alpha = \frac{2510-747s}{8640}, \qquad \omega = \frac{7773-2447s}{8640}. \end{gathered} \tag{4.1}$$

The letters p, q, t in this section without an argument denote these fixed limiting values. For a variable Bernoulli parameter v, use $a(v) = v(1-v)$, $r(v) = v^2 + (1-v)^2$ and $\tau(v) = 1-2v$. For a real number x, write $\langle x\rangle = x - \mathrm{round}(x)$, choosing either convention at half-integers.

Theorem 4.1 (Eventual binomial padding)**.** *At every sufficiently large horizon N, P_N has a unique maximizing core size m_N, and $j_N = N - m_N$ belongs to $\{0,1,2,3,4\}$. The limiting frequencies of these five zero counts are respectively*

$$p-\omega, \quad p-\omega, \quad t, \quad \omega, \quad \omega.$$

The gaps between successive record core sizes eventually belong exactly to $\{2,3,5\}$: no other gap occurs sufficiently late, and each of these three occurs infinitely often. No effective first horizon is asserted.

4.1 A uniform expansion at a fixed integer threshold

Let $F_{m,v}$ be the distribution function of $\mathrm{Bin}(m,v)$. Fix an integer k and set $c = mv - k$, $d = (ma(v))^{-1/2}$. The positive closed-threshold discrepancy is $D_{m,k}(v) = F_{m,v}(k) - \Phi(-cd)$.

Lemma 4.2. *On every compact $J \subset (0,1)$ and bounded range $|c| \le M$, uniformly for $v = (k+c)/m \in J$,*

$$D_{m,k}(v) = \phi(0)\big[d(1/2 + \tau/6) + d^3 C(\tau,-c) + d^5 T_5(\tau,-c)\big] + O(d^7), \tag{4.2}$$

with the same order for the derivative of the remainder in c, fixing m,k. The expansion extends to

any fixed order and any fixed finite number of these branch derivatives. Here

$$C(\tau,\beta)=\frac{-1080(1+\tau)\beta^2-(90\tau^2+1080\tau+630)\beta+\tau^3-45\tau^2-189\tau-135}{4320},$$
$$\begin{aligned}T_5(\tau,\beta)=&\frac{3+5\tau}{48}\beta^4+\frac{27+60\tau+25\tau^2}{288}\beta^3+\frac{15+31\tau+25\tau^2+\tau^3}{192}\beta^2\\&+\frac{549+1320\tau+1030\tau^2+120\tau^3+5\tau^4}{23040}\beta\\&+\frac{189+927\tau+126\tau^2+234\tau^3+21\tau^4-25\tau^5}{193536}.\end{aligned}$$

Proof. The Fourier integral for a binomial mass is on $[-\pi,\pi]$. Its modulus satisfies

$$|1-v+ve^{i\xi}|^m=(1-4a(v)\sin^2(\xi/2))^{m/2}.$$

After $z=\xi\sqrt{ma(v)}$, the inequality $\sin(|\xi|/2)\geq|\xi|/\pi$ bounds this by e^{-2z^2/π^2} throughout the integration interval. On J, the centered logarithm has a Taylor expansion of any prescribed order with uniformly bounded coefficients and parameter derivatives. Split the integral at $|z|=m^{\varepsilon}$, with ε sufficiently small for that order. Polynomial multiples of the Gaussian bound control the inner Taylor remainder; the outer integral is smaller than any prescribed power of m^{-1}. Taking sufficiently many terms gives a Hermite mass expansion with uniform absolute error $O(d^{12})$, or any higher fixed power if needed, at every integer mass index.

This statement also holds for branch derivatives. Before centering,

$$\frac{\partial}{\partial c}(1-v+ve^{i\xi})^m=(e^{i\xi}-1)(1-v+ve^{i\xi})^{m-1},\qquad v=(k+c)/m.$$

Further fixed derivatives have the same Gaussian control after increasing the retained Taylor order. The standardized mass index and spacing have bounded branch derivatives of the needed orders on the stated ranges.

Summing at most $m+1=O(d^{-2})$ masses turns the displayed mass error into $O(d^{10})$, including its branch derivatives. The tails beyond the true binomial support are exponentially small. Euler–Maclaurin summation may therefore be applied to the infinite lower lattice of the Gaussian-polynomial terms. Its order-eight remainder is bounded by a constant times $d^8\|f^{(8)}\|_1$ for each such term. In the integral periodic-Bernoulli remainder, differentiating the phase gives bounded factors of the form $1+d|z|$; Gaussian derivative moments remain integrable. Increasing the summation order gives the corresponding assertion for further branch derivatives.

The coefficient of d^j in the density has parity j. Integration, the half-cell term, and the Euler–Maclaurin derivative terms then leave only odd powers $d, d^3, d^5, \dots$ when evaluated at $-cd$. The terms through d^5 give (4.2). Their coefficients follow by using the Bernoulli cumulant ratios

$$\tau,\quad 1-6a,\quad \tau(1-12a),\quad 1-30a+120a^2,\quad \tau(1-60a+360a^2)$$

and the lattice corrections $df/2+d^2f'/12-d^4f'''/720$. The cubic phase coefficient is also the polynomial in Schulz's Remark 4.3, derived on p. 25 [8]. □

Dividing (4.2) by $r(v)d$ gives the threshold efficiency

$$R_{m,+,k}(v)=E(v)-\frac{Q(v,c)}{m}-\frac{H(v,c)}{m^2}+O(m^{-3}),\tag{4.3}$$

where

$$E(v)=\phi(0)\frac{3+\tau(v)}{3(1+\tau(v)^2)},\quad Q(v,c)=-\frac{\phi(0)C(\tau(v),-c)}{r(v)a(v)},\quad H(v,c)=-\frac{\phi(0)T_5(\tau(v),-c)}{r(v)a(v)^2}.$$

The phase parabola $Q(v,c)$ has vertex

$$c_0(v)=\frac{\tau(v)^2+12\tau(v)+7}{24(1+\tau(v))}$$

and minimum

$$Q_{\min}(v)=\frac{\phi(0)\{345+72\tau-498\tau^2-8\tau^3-23\tau^4\}}{4320(1+\tau)(1-\tau^4)}. \tag{4.4}$$

4.2 Global binomial localization and the padding envelope

Schulz's Theorem 1 gives $B_m < C_{\mathrm{E}}$ for every finite m. His Proposition 6.1 bounds the full Kolmogorov distance on $v \in [1/6, 5/6]$ by the four absolute closed/left central-threshold errors at $\lfloor mv \rfloor, \lceil mv \rceil$ and

$$\gamma\frac{\phi(0)(3+|1-2v|)}{6\sqrt{mv(1-v)}},\qquad \gamma=\frac{2}{\sqrt{2\pi}C_{\mathrm{E}}}-1=12t-1<1.$$

His estimate on the two complementary parameter ranges is $R_m(v) \le .3031(1+.646/\beta(v)) < .4045$, where $\beta(v)=r(v)/\sqrt{a(v)}$ [8, pp. 29–31]. Each B_m is attained: the ratio is the maximum of finitely many continuous threshold expressions and tends to zero as $v \to 0, 1$.

By reflection restrict $v \le 1/2$. Central expansions and the source bound show uniform convergence of the optimized threshold value to $E(v)$ on the retained compact parameter interval. Its unique maximum is C_{E} at $v=p$. Thus every maximizing parameter tends to p. Near p, the extra γ term and left-limit thresholds are uniformly smaller than the positive closed central threshold: the latter has leading coefficient $\phi(0)(1/2+\tau/6)/r$, while the left-limit coefficient is $\phi(0)(1/2-\tau/6)/r$. Consequently the full winner is eventually a positive closed floor/ceiling threshold, with $|c| \le 1$.

Taking $v=(k+c_*)/m$ with k nearest $mp-c_*$ gives $C_{\mathrm{E}}-Q_*/m+O(m^{-2})$. Since $E(v)=C_{\mathrm{E}}-\alpha(v-p)^2+O((v-p)^3)$, comparison with (4.3) and $Q \ge Q_{\min}$ yields

$$(v-p)^2 \le C|v-p|/m+C/m^2,\qquad v-p=O(m^{-1}).$$

The positive quadratic phase penalty then gives $c-c_0(v)=O(m^{-1/2})$. Since $0<c_*<1$, the branch is eventually interior. Differentiating (4.3) along $v=(k+c)/m$ and using $E'(v)=O(m^{-1})$ gives $Q_c=O(m^{-1})$, so $c-c_0(v)=O(m^{-1})$ and $v=(k+c_*)/m+O(m^{-2})$.

Set $e=k+c_*-mp$. Taylor expansion and completion of the square give

$$B_m=C_{\mathrm{E}}-\frac{Q_*}{m}-\frac{H_*+\alpha\,\mathrm{dist}(mp-\delta,\mathbb{Z})^2}{m^2}+O(m^{-3}). \tag{4.5}$$

Indeed the second-order branch correction is $\alpha e^2+Q_p e+H_0$, where all derivatives are evaluated at (p,c_*), $H_0=H(p,c_*)$, $b=-Q_p/(2\alpha)$ and $H_*=H_0-\alpha b^2$. Direct substitution in (4.4) gives (4.1). The formula remains uniform at the nearest-integer crossing, by taking the better of the two branch values; the nearest branch itself need only be selected outside an $O(m^{-1})$ half-integer neighborhood.

Choose the most recent maximizing core m_N in a tie. Strict finite $B_m < C_{\mathrm{E}}$ and $B_m \to C_{\mathrm{E}}$ imply $m_N \to \infty$. The leading correction in (4.5), compared with B_N, then gives $m_N/N \to 1$, and

its $O(m^{-2})$ bounds give $j_N = N - m_N = O(1)$. With $\theta_N = \mathrm{round}(Np - \delta) - (Np - \delta)$, for bounded j,

$$N^2(B_{N-j} - B_N) = \alpha\{\theta_N^2 - \mathrm{dist}(\theta_N + jp, \mathbb{Z})^2 - \eta j\} + O(N^{-1}).$$

The limiting policy minimizes $f_j(\theta) = \eta j + \mathrm{dist}(\theta + jp, \mathbb{Z})^2$. Define

$$\begin{aligned} T_{13} &= -(p+u)/2 + \eta/t, & T_{30} &= -u/2 - 3\eta/(2u), \\ T_{02} &= t/2 + \eta/t, & T_{24} &= 3t/2 + \eta/t, \\ T_{41} &= (q+2t)/2 - 3\eta/(2u). \end{aligned} \tag{4.6}$$

In increasing order on $[-1/2, 1/2]$ the winning zero counts are

Phase interval	Zero count
$[-1/2, T_{13}]$	1
$[T_{13}, T_{30}]$	3
$[T_{30}, T_{02}]$	0
$[T_{02}, T_{24}]$	2
$[T_{24}, T_{41}]$	4
$[T_{41}, 1/2]$	1

The relevant parabola centers are $-p, -u, 0, t, 2t, q$, with biases $\eta, 3\eta, 0, 2\eta, 4\eta, \eta$. Subtracting neighboring quadratics gives exactly (4.6). The remaining copies are above this envelope: their differences are affine, so it suffices to check the displayed interval endpoints. The successive active widths are $\omega, p - \omega, t, \omega$, with the two outer pieces totaling $p - \omega$. Positivity follows, in particular, from $7773^2 - 10(2447)^2 = 541439 > 0$.

For completeness all $j \geq 5$ have a uniform positive price gap. Using only $j = 0, 1, 2$, the relevant centers are $-p, 0, t, q$ and the intersections are $-p/2 - \eta/(2p)$, $t/2 + \eta/t$ and $(q+t)/2 - \eta/(2p)$. The largest potentially troublesome intersection price is at most $2\eta + p^2/4 < 5\eta$, with gap greater than $59/10000$, using

$$4/25 < t < 1/6, \qquad 5/12 < p < 21/50, \qquad 1/60 < \eta < 1/55.$$

The other intersection prices are below $1/16 < 5\eta$; the endpoint price $\eta + t^2/4$ is also below 5η. But $f_j \geq 5\eta$ for $j \geq 5$. The bounded-j comparison therefore gives $j_N \leq 4$ eventually.

Away from $O(N^{-1})$ neighborhoods of the five crossings, the actual winner has the displayed code. Since p is irrational, its rotation is equidistributed. Those shrinking neighborhoods have density zero: their upper density is bounded by the length of any fixed larger neighborhood. The interval widths prove the claimed padding frequencies.

4.3 Boundary drift and exact eventual record gaps

To decide every sufficiently late crossing, extend Lemma 4.2 to the seventh CDF order. Its smooth efficiency expansion has remainder $O(m^{-4})$, including the branch derivatives needed for optimization. The seventh-order coefficient's value at (p, c_*) will be denoted J_*. Its value is common to the core comparisons and will cancel.

All derivatives below are of the explicit Q, H above at (p, c_*). Put

$$\begin{aligned} K(e) &= (2\alpha + Q_{pc})e + Q_p + H_c, & F_2(e) &= \alpha e^2 + Q_p e + H_0, \\ G(e) &= E_3 e^3 - Q_{pp} e^2/2 - H_p e + K(e)^2/(2Q_{cc}). \end{aligned}$$

The needed constants are

$$Q_{cc} = \phi(0)(35 + 11s)/15, \quad Q_{pc} = \phi(0)(415 + 131s)/540, \quad Q_p = \phi(0)(1483 + 463s)/19440,$$
$$E_3 = E'''(p)/6 = -\phi(0)(19 + 6s)/15, \quad c_0'(p) = -(23 + 4s)/108, \quad Q_{pp} = Q''_{\min}(p) + Q_{cc}c_0'(p)^2.$$

The derivatives H_c, H_p are obtained by differentiating the displayed T_5 expression for H; thus every coefficient here is explicit. Writing $c = c_* + d_1/m$ and $v = p + e/m + d_1/m^2$, branch stationarity gives $d_1 = -K(e)/Q_{cc}$. Substituting into the branch value yields

$$B_m = C_{\mathrm{E}} - Q_*/m - F_2(e)/m^2 + (G(e) - J_*)/m^3 + O(m^{-4}) \tag{4.7}$$

on a winning integer branch. Near a core's internal half-integer crossing both branches must be compared. At each policy crossing in (4.6), the two competing cores' internal phases lie in $[T_{30}, T_{02}] \subset (-1/2, 1/2)$, so there is a fixed distance from those internal crossings.

For $\theta_j = \langle \theta + jp \rangle$ and $e_j = b + \theta_j$, put

$$L_j(\theta) = Q_* j + F_2(e_j), \qquad M_j(\theta) = -Q_* j^2 - 2jF_2(e_j) + G(e_j).$$

Expanding (4.7) at $m = N - j$ gives

$$B_{N-j} = C_{\mathrm{E}} - Q_*/N - L_j(\theta)/N^2 + (M_j(\theta) - J_*)/N^3 + O(N^{-4}).$$

At a transverse limiting crossing T between i, j, the price difference changes sign at

$$\theta = T + \frac{D_{ij}}{N} + O(N^{-2}), \qquad D_{ij} = \frac{M_i(T) - M_j(T)}{L_i'(T) - L_j'(T)}.$$

Here the equation refers to the uniform sign comparison of branch values; no continuous interpolation of integer-threshold optima is required. Substitution of the explicit polynomials gives

$$\begin{aligned} D_{13} = D_{02} = D_{24} = D_A &= \frac{129918319}{529079040} + \frac{191267395}{1693052928}s, \\ D_{30} = D_{41} = D_B &= -\frac{663979357}{2351462400} - \frac{20072514361}{141087744000}s. \end{aligned} \tag{4.8}$$

Both rational parts are nonzero, so neither sD_A nor sD_B is rational.

The actual rotation cannot remain ambiguously close to a shifted boundary. Let $h_N = \theta_N - T$ and $\gamma = \delta - T$. With $L_0 = 25920$, the numerators of $\gamma = (\gamma_A + \gamma_B s)/L_0$ are

T	γ_A	γ_B
T_{13}	95720	-25573
T_{30}	72401	-18232
T_{02}	43880	-12613
T_{24}	121640	-38533
T_{41}	98321	-31192

These follow directly from $\eta/t = (60 + 269s)/8640$ and $3\eta/(2u) = (269 + 6s)/2880$. If k is the nearest integer to $Np - \delta$, then

$$h_N = (A_N + B_N s)/L_0, \quad A_N = L_0(k - 2N) + \gamma_A, \quad B_N = 12960N + \gamma_B.$$

Its field norm is

$$\nu_N = (A_N^2 - 10B_N^2)/L_0^2 \in L_0^{-2}\mathbb{Z}.$$

In any fixed corridor $|h_N| \le C/N$, the conjugate of h_N is $-sN + O(1)$, so ν_N is bounded and takes only finitely many rational values. Moreover $Nh_N = -\nu_N/s + O(N^{-1})$. For the corresponding D_A or D_B, each $-\nu/s - D$ is nonzero by (4.8). The finite set has a positive minimum separation. Thus $|\theta_N - T - D/N| \ge \varepsilon/N$ eventually for some $\varepsilon > 0$. Outside a sufficiently wide C/N corridor the same separation is immediate. This margin dominates the $O(N^{-2})$ boundary uncertainty. Other policies have fixed limiting price gaps near each crossing; away from crossings the unique leading winner persists. Hence the exact sufficiently late code is the six-interval table, shifting the three A endpoints by D_A/N and the two B endpoints by D_B/N. In particular the winning core is unique.

Finally the endpoint translations

$$T_{02} - p = T_{13}, \qquad T_{13} + q = T_{24}, \qquad T_{30} + q = T_{41}$$

preserve the drift group. Changing N to $N + 1$ changes D/N only by $O(N^{-2})$, below the norm separation. Interval transport under $\theta_{N+1} = \langle \theta_N - p \rangle$ therefore gives exactly

$$0 \longrightarrow 1, \quad 1 \longrightarrow 0 \text{ or } 2, \quad 2 \longrightarrow 0 \text{ or } 3, \quad 3 \longrightarrow 4, \quad 4 \longrightarrow 0.$$

The return lengths to a record are $2, 3, 5$. Their frequencies per horizon are $u - \omega, t - \omega, \omega$, respectively, all positive; this follows by translating the displayed intervals, with the vanishing shifts leaving frequencies unchanged. Their sum is $p - \omega$, and their gap-weighted sum is one. Each gap thus occurs infinitely often, proving Theorem 4.1.

5 Exact replacement of nearby diameters

The comparator preserves the individual probability vector. Unequal probabilities make the conditional law given the count nonuniform; small probabilities also change the scale of the possible improvement. We keep weighted count moments throughout the proof and condition only after locating a competitive threshold.

Scale by h_0 from Theorem 1.3, and put

$$q_i = 1 - p_i, \quad r_i = p_i^2 + q_i^2, \quad R_0 = \sum_i a_i r_i, \quad \mu = \sum_i p_i, \quad \ell = \sum_i a_i |e_i|.$$

Then $\sum_i a_i e_i = 0$, $A/2 \le R_0 \le A$ and $\ell \le \sqrt{A}\sigma$. With

$$K = \sum_i B_i, \qquad T = \sum_i e_i (B_i - p_i),$$

the original centered sum is $K - \mu + T$, and its variance is $A + \sigma^2$. Write $c_k = \mathbb{P}(K = k)$ and $N_j(k) = \mathbb{E}[T^j \mathbf{1}_{\{K=k\}}]$.

5.1 Count estimates with no probability floor

Uniformly over all individual probabilities and every integer k, as $A \to \infty$,

$$\begin{aligned} c_k &= \frac{\phi(x_k)}{\sqrt{A}} + O(A^{-1}), \\ F_K(k) - \Phi(x_k) &= \frac{\phi(x_k)}{\sqrt{A}} \left\{ \frac{1}{2} + \frac{\gamma_3}{6}(1 - x_k^2) \right\} + O(A^{-1}), \\ x_k &= (k - \mu)/\sqrt{A}, \qquad \gamma_3 = A^{-1} \sum_i a_i (1 - 2p_i), \quad |\gamma_3| \le 1. \end{aligned} \tag{5.1}$$

All constants here are absolute. To prove the mass formula, use Fourier inversion, the local cumulant expansion, and

$$\left|\prod_{i\notin F}(q_i+p_ie^{it})\right| \le \exp\left\{-2\left(A-\sum_{i\in F}a_i\right)\sin^2(t/2)\right\}.$$

For at most four deletions, $\sum_{i\in F}a_i \le 1$. Every fixed-order Bernoulli cumulant of order at least two is bounded by a constant times a_i. The local Taylor remainders have the same bound because $q_i+p_ie^{it}$ stays uniformly away from zero near zero. Thus the Fourier integral of the error from the leading Gaussian is $O(A^{-1})$, independently of k.

For the CDF introduce independent $U \sim \mathrm{Unif}[-1/2,1/2]$. The exact identity is $F_K(k) = F_{K+U}(k+1/2)$. Near zero, comparison of the characteristic function with

$$e^{-At^2/2}\{1+\kappa_3(it)^3/6\}, \qquad \kappa_3 = A\gamma_3,$$

has CDF-inversion error bounded by a constant times

$$\int (A|t|^3 + A^2|t|^5 + t^2/|t|)e^{-cAt^2}\,dt = O(A^{-1}).$$

The last term is the error from the jitter factor $\mathrm{sinc}(t/2)$. At every nonzero peak $t = 2\pi j + u$, $|u| \le \pi$,

$$|\,\mathrm{sinc}(t/2)/t| \le C|u|/j^2.$$

The product modulus therefore bounds that peak by $O(1/(Aj^2))$. Summing these bounds and using the Gaussian tails proves the jittered CDF expansion uniformly in the threshold. Expanding its half-grid shift gives (5.1). This proof uses a continuous jittered CDF and actual integer-grid values of F_K.

The required weighted moments satisfy, uniformly for every k,

$$\begin{aligned} |N_1(k)| &\le C\ell A^{-3/2}, \\ |N_2(k) - \sigma^2 c_k| &\le C\sigma^2/A, \\ 0 \le N_4(k) &\le C(\sigma^4+\eta^2\sigma^2)/\sqrt{A}. \end{aligned} \tag{5.2}$$

Here is a direct Fourier proof retaining the vanishing spread factors. Put $f_i = q_i + p_ie^{it}$, $s = e^{it}-1$, $\Psi = \prod f_i$, and let Ψ_i, Ψ_{ij} denote deleted products. For $Y_i = B_i - p_i$,

$$\mathbb{E}[Y_ie^{itB_i}] = a_is, \qquad \mathbb{E}[Y_i^2e^{itB_i}] = a_if_i + (1-2p_i)a_is, \qquad |\mathbb{E}[Y_i^je^{itB_i}]| \le a_i \quad (j\ge 2).$$

Weighted orthogonality gives the exact cancellation

$$\widehat{N}_1 = s\sum_i a_ie_i\Psi_i = -s^2\sum_i a_ip_ie_i\Psi_i.$$

Its absolute integral is $O(\ell A^{-3/2})$. For the second moment, subtracting $\sigma^2\Psi$ leaves

$$s\sum_i e_i^2(1-2p_i)a_i\Psi_i + s^2\sum_{i\ne j}u_iu_j\Psi_{ij}, \qquad u_i = a_ie_i, \quad \sum_i u_i = 0.$$

Use the exact identity

$$\Psi_{ij} = \Psi + (1-f_i)\Psi_i + (1-f_j)\Psi_j + (1-f_i)(1-f_j)\Psi_{ij}.$$

The cross term is bounded by $C\sigma^2(t^2+|t|^3+At^4)e^{-cAt^2}$; the singleton term has integral $O(\sigma^2/A)$. This proves the second estimate without dividing by an individual characteristic function. For the fourth moment, the index partitions $4, 31, 22, 211, 1111$ and

$$\sum_i a_i e_i^4 \le \eta^2\sigma^2, \quad \sum_i a_i|e_i|^3 \le \eta\sigma^2, \quad \ell \le \sqrt{A}\sigma$$

give respective absolute integral bounds controlled by $C(\sigma^4+\eta^2\sigma^2)/\sqrt{A}$. For example, the 31 term is at most $C\eta\sigma^3/\sqrt{A}$; $2\eta\sigma^3 \le \sigma^4+\eta^2\sigma^2$ absorbs it. Fourier inversion proves (5.2), including rare layers and layers of probability zero.

5.2 Locating thresholds before conditioning

At an actual positive closed threshold y, write $y = k-\mu+\theta$, with k nearest to $y+\mu$ and $|\theta| \le 1/2$. The exact decomposition is

$$F_h(y) = F_K(k) - c_k\mathbb{P}(T > \theta \mid K = k) + E_{k,\theta}, \quad |E_{k,\theta}| \le \frac{C(\sigma^4+\eta^2\sigma^2)}{\sqrt{A}}. \tag{5.3}$$

When $c_k = 0$ the subtracted product is zero. To bound the remainder, sum $N_4(j)/(|j-k|-1/2)^4$ over all $j \ne k$. This retains all interleaving count layers and the closed convention.

The original denominator is

$$R_h - R_0 = 3\sum_i a_i r_i e_i + 3\sum_i a_i r_i e_i^2 + \sum_i a_i r_i e_i^3, \quad |R_h - R_0| \le C(\ell+\sigma^2+\eta\sigma^2). \tag{5.4}$$

Consequently, for $P_h = (A+\sigma^2)^{3/2}/R_h$ and $P_0 = A^{3/2}/R_0$, $P_h/P_0 = 1+O(\ell/A+\sigma^2/A)$ when σ, η are small. Dropping the nonpositive term in (5.3), using (5.1) and globally bounded normal derivatives, first gives every positive discrepancy as $O(A^{-1/2})$. The corresponding global ratio bound is

$$\mathcal{C}_h(y) \le C(1+x_k^2)\phi(x_k) + C(\sigma^4+\eta^2\sigma^2) + O((1+\sigma)/\sqrt{A}).$$

Choose the spread bounds small and then A large, depending on c_0. Any active threshold with ratio at least c_0 has bounded x_k. Thus $c_k \ge c_{c_0}/\sqrt{A}$ and

$$\begin{aligned} &|\mathbb{E}(T \mid K = k)| \le C\ell/A, \\ &\mathbb{E}(T^2 \mid K = k) = \sigma^2\{1+O(A^{-1/2})\}, \\ &\mathbb{E}(T^4 \mid K = k) \le C(\sigma^4+\eta^2\sigma^2). \end{aligned} \tag{5.5}$$

Only at this stage have we divided the weighted estimates by c_k.

5.3 An absolute-moment bound from two conditional samples

At the localized count, we need the bound

$$\mathbb{E}(|T| \mid K = k) \ge c\ell/\sqrt{A}. \tag{5.6}$$

The count-conditioned law is the classical rejective sampling law [3, Section 2]. The relative estimates used here follow directly from our grid formula. For at most three fixed coordinates F, direct factorization gives

$$\mathbb{P}(B_F = b_F \mid K = k) = \mathbb{P}(B_F = b_F)\frac{c_{-F}(k-\sum b_F)}{c_k}.$$

The deleted variance differs by at most $3/4$ and the standardized argument by $O(A^{-1/2})$. The mass estimate in (5.1) therefore makes the ratio $1 + O(A^{-1/2})$, uniformly even when the configuration probability tends to zero.

Take two independent copies B, B' of the count-conditioned law. Let $C = \{i : B_i \neq B'_i\}$ and $m = |C|$, an even integer. Given C and the common successes outside C, the change signs are uniform balanced signs: the product of their odds has one factor p_i/q_i for every changed coordinate, independently of the sign. Generate these signs by a uniform perfect matching of C and independent opposite Rademacher signs in each pair. For a fixed matching, moment interpolation gives mean absolute value at least $1/\sqrt{3}$ times the Euclidean norm of its coefficients $e_i - e_j$. Averaging matchings gives, for $m \geq 2$,

$$\mathbb{E}(|T - T'| \mid C) \geq \sqrt{2/3}\, \frac{\sum_{i<j \in C} |e_i - e_j|}{(m-1)\sqrt{m}}.$$

The relative deletion ratios imply

$$\mathbb{P}(i, j \in C) \geq c a_i a_j, \qquad \mathbb{E}(m \mid i, j \in C) \leq CA.$$

For the first identity, if π_{ab} are the conditional two-coordinate state probabilities, the exact probability is $2(\pi_{11}\pi_{00} + \pi_{10}\pi_{01}) = 4a_i a_j\{1 + O(A^{-1/2})\}$. For the second, fix the two change signs. This fixes two coordinates and therefore the remaining count in each of the two independent conditional samples. The one-coordinate relative deletion ratios in these reduced samples bound the probability of a difference at coordinate l by Ca_l. Summing gives $\mathbb{E}(m \mid i, j \in C) \leq 2 + CA = O(A)$. Jensen's inequality for $m^{-3/2}$ then yields

$$\mathbb{E}|T - T'| \geq cA^{-3/2} \sum_{i<j} a_i a_j |e_i - e_j|.$$

Orthogonality gives $\sum_j a_j |e_i - e_j| \geq A|e_i|$, so the sum is at least $A\ell/2$. The triangle inequality $\mathbb{E}|T - T'| \leq 2\mathbb{E}|T|$ proves (5.6).

5.4 The phase loss and the exact final comparison

Write $M_1 = \mathbb{E}(|T| \mid K = k)$, $W = \mathbb{P}(T > \theta \mid K = k)$ and $L = \theta + W$. Equations (5.5)–(5.6) give

$$|\mathbb{E}(T \mid K = k)| \leq M_1/4, \qquad M_1 \geq c\frac{\sigma^2}{\sigma + \eta}$$

for A large. The second inequality is interpolation between the second and fourth moments. Thus $\mathbb{E}(T^+ \mid K = k) \geq 3M_1/8$. Cauchy's inequality gives

$$\nu := \mathbb{P}(T > 0 \mid K = k) \geq cM_1^2/\sigma^2 \geq cM_1/(\sigma + \eta).$$

For small enough $\sigma + \eta$, $\nu \geq 8M_1$. If $\theta \geq M_1/4$, then $L \geq M_1/4$. For $0 \leq \theta < M_1/4$, removing the lower positive values leaves positive first moment at least $M_1/8$; Cauchy's inequality again gives $W \geq cM_1^2/\sigma^2 \geq cM_1$. If $\theta = -u < 0$ and $u \leq \nu/2$, then $L \geq \nu/2$. If $u > \nu/2 \geq 4M_1$, first-moment Markov gives $\mathbb{P}(T \leq -u \mid K = k) \leq 1/4$, hence $L \geq 1/4$. These cases prove the relative bounds

$$L \geq c \max\left(\frac{\ell}{\sqrt{A}}, \frac{\sigma^2}{\sigma + \eta}\right) > 0, \quad |\theta| + \theta^2 + W \leq CL, \quad \sigma^2 \leq C(\sigma + \eta)L. \tag{5.7}$$

Compare with the exact unit-diameter closed discrepancy $D_0(k) = F_K(k) - \Phi(x_k)$, and set $\alpha_k = \phi(x_k)/\sqrt{A}$. At the bounded standardized threshold, normal Taylor expansion gives

$$D_h(y) - D_0(k) = -c_k L + (c_k - \alpha_k)\theta + E_{k,\theta} + O((\sigma^2 + \theta^2)/A).$$

Multiply by P_0, and use (5.4) to include $(P_h - P_0)D_h$. Every final error contains L: the normal, mass and denominator terms are $O(L/\sqrt{A})$, and the crossing term is at most $C(\sigma+\eta)^3 L$. The leading coefficient $P_0 c_k$ has a positive lower bound. Consequently

$$\mathcal{C}_h(y) - P_0 D_0(k) \le \{-c + C(\sigma+\eta)^3 + C/\sqrt{A}\}L \le -\kappa \frac{\sigma^2}{\sigma+\eta}.$$

Choose the spread bounds and A_0 in that order. The comparison uses the exact common-diameter branch, so no spread-independent grid error remains. Its branch lies below the full common-diameter ratio. Reflection of a negative strict maximizing threshold preserves a_i, r_i, σ, η and treats the other side. This proves Theorem 1.3. If $p_i \in J \Subset (0,1)$, then $\eta \le \sigma/\sqrt{\min_{p\in J} p(1-p)}$, giving its linear-$\sigma$ corollary.

5.5 Why the probability-free gain has the smaller scale

Fix $c_0 < C_{\mathrm{E}}$. Choose a fixed, sufficiently large IID unit-diameter binary group of size m with $p = p_{\mathrm{E}}$, ratio above c_0, and a positive closed maximum strictly above its negative strict maximum. The phase expansion in Section 4 supplies such groups, with m above any prescribed count-variance cutoff. Put $a = p(1-p)$, $r = p^2 + (1-p)^2$. Add a Bernoulli(s) coordinate. For small s, its common-diameter maximum still has the positive closed sign, by continuity of the finite set of threshold branches. Denote a winning integer by $k(s)$.

Give the group diameter $g_c = 1 + s(1-s)\eta/(ma)$ and the rare coordinate diameter $g_s = 1 - \eta$, with $0 < \eta < 1/4$. This preserves the weighted mean diameter one. At raw threshold kg_c the CDF is exactly

$$(1-s)F_{\mathrm{group}}(k) + sF_{\mathrm{group}}(k-1),$$

the same as the common-diameter array, since $0 < g_s/g_c < 1$. The centered threshold, variance and third-moment sum are

$$y_k = kg_c - mpg_c - sg_s, \quad V_h = mag_c^2 + s(1-s)g_s^2, \quad R_h = marg_c^3 + s(1-s)\{s^2 + (1-s)^2\}g_s^3.$$

For fixed m, p, the corresponding finitely many smooth branch formulas are analytic near $(s,\eta) = (0,0)$. They agree with the common-diameter branch at $\eta = 0$ and are independent of η at $s = 0$. A bounded mixed derivative therefore gives

$$0 < \mathcal{C}_{\mathrm{com}}(s) - \mathcal{C}_h \le K_m s\eta,$$

where the lower strictness follows from Theorem 1.3. Meanwhile $\sigma^2 = s(1-s)\eta^2\{1 + s(1-s)/(ma)\}$. Taking $s = \eta^4$ gives $\sigma \sim \eta^3$ and a gain of order η^5, equal to the order $\sigma^2/(\sigma+\eta)$ supplied by the theorem. These arrays stay above c_0 by continuity. Thus a uniform gain proportional to σ fails without a probability floor, even for competitive finite binary arrays.

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